Q30: Discussion Questions

DLS 2026 — Kukla

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1. TorchDynamo inserts a "graph break" when it hits untraceable Python. What kinds of code cause graph breaks, and why is graceful fallback better than refusing to compile?
2. AOTAutograd captures the backward graph ahead of time. What optimizations become possible once the compiler can see forward and backward together?
3. Activation memory, not parameter memory, usually dominates training. Walk through why, for a deep network with a large batch.
4. Gradient checkpointing trades compute for memory. How would you choose *which* layers to checkpoint to get the best trade?
5. Functional AD (JAX-style) is elegant and composable but can be memory-hungry if naive. Is the right fix better compilers or better programmer discipline? Argue.